Prompt 1 — Create the basic lab application

Create a simple educational web application for an undergraduate AIML laboratory.

Title:

Discrete Probability Distributions

The application will eventually contain:

1. Binomial Distribution

2. Poisson Distribution

3. Geometric Distribution

For now, implement only the basic application structure and the Binomial Distribution module.

Keep the UI simple and clean. Do not add unnecessary features.

Create a navigation/header with:

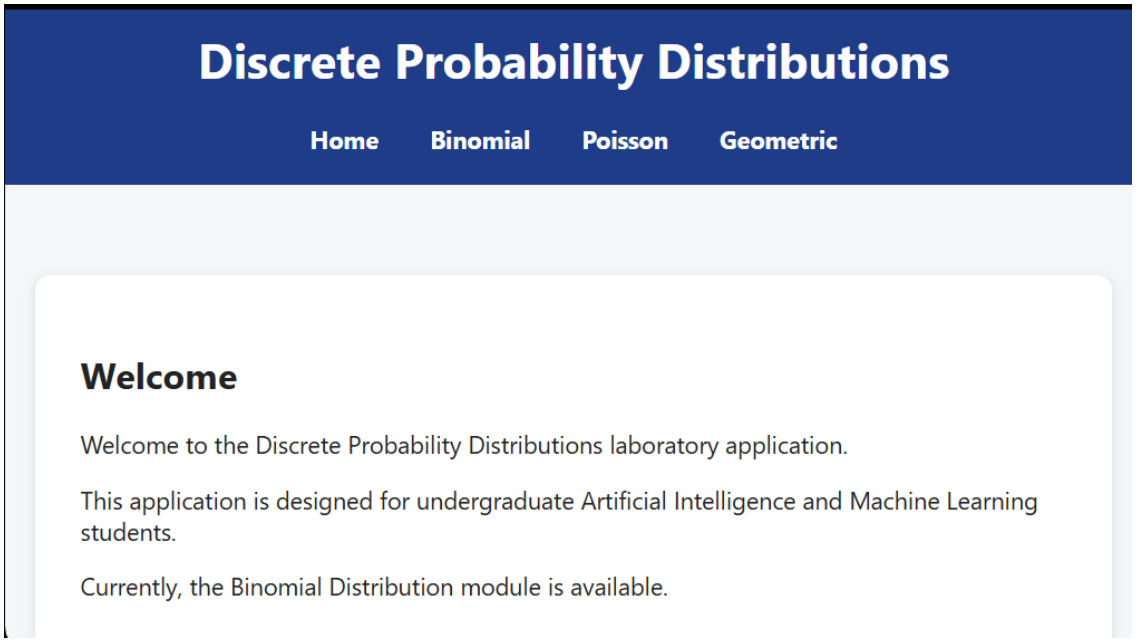
- Home

- Binomial

- Poisson

- Geometric

Only the Binomial page needs to be functional at this stage.



Binomial Distribution

Prompt 2 — Add Binomial inputs

Implement the Binomial Distribution section.

Add two input controls:

- Number of trials n

- Probability of success p

Set default values:

n = 20

p = 0.35

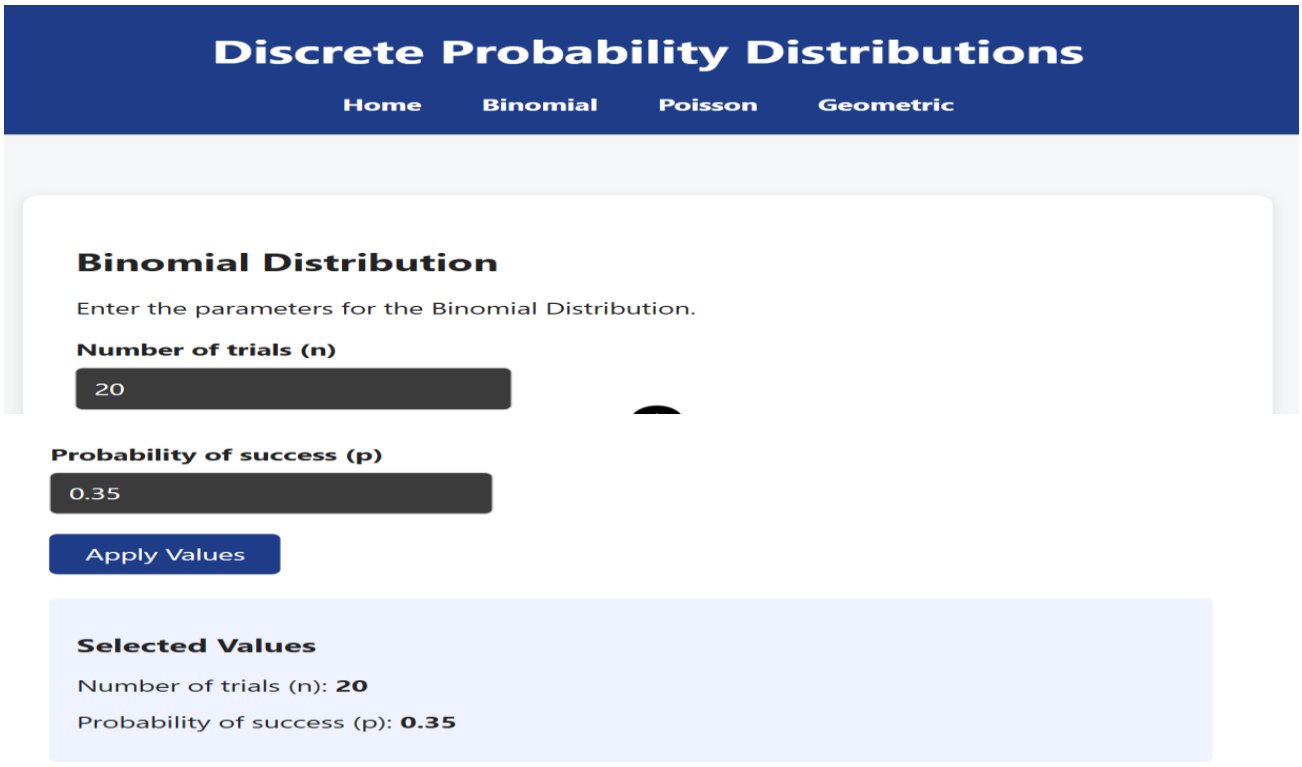
Validate the inputs:

- n must be a positive integer

- p must be between 0 and 1

Display the selected values clearly.

Do not add simulation yet.



Prompt 3 — Calculate Binomial statistics

Add the Binomial probability calculations using the selected n and p.

Calculate and display:

- Mean

- Variance

- Standard deviation

- Second moment $E[X^2]$

- Skewness

Use the following formulas:

Mean:

np

Variance:

np(1-p)

Second moment:

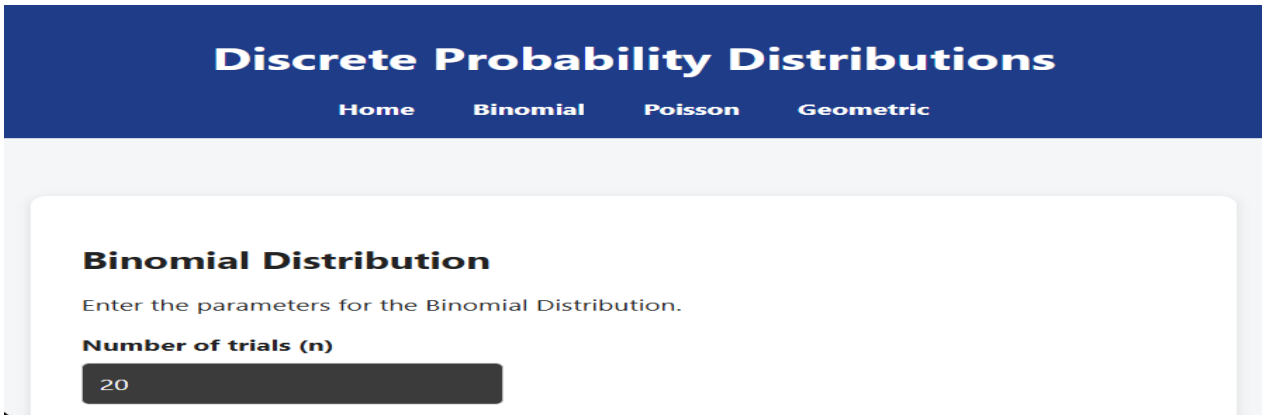
Variance + Mean²

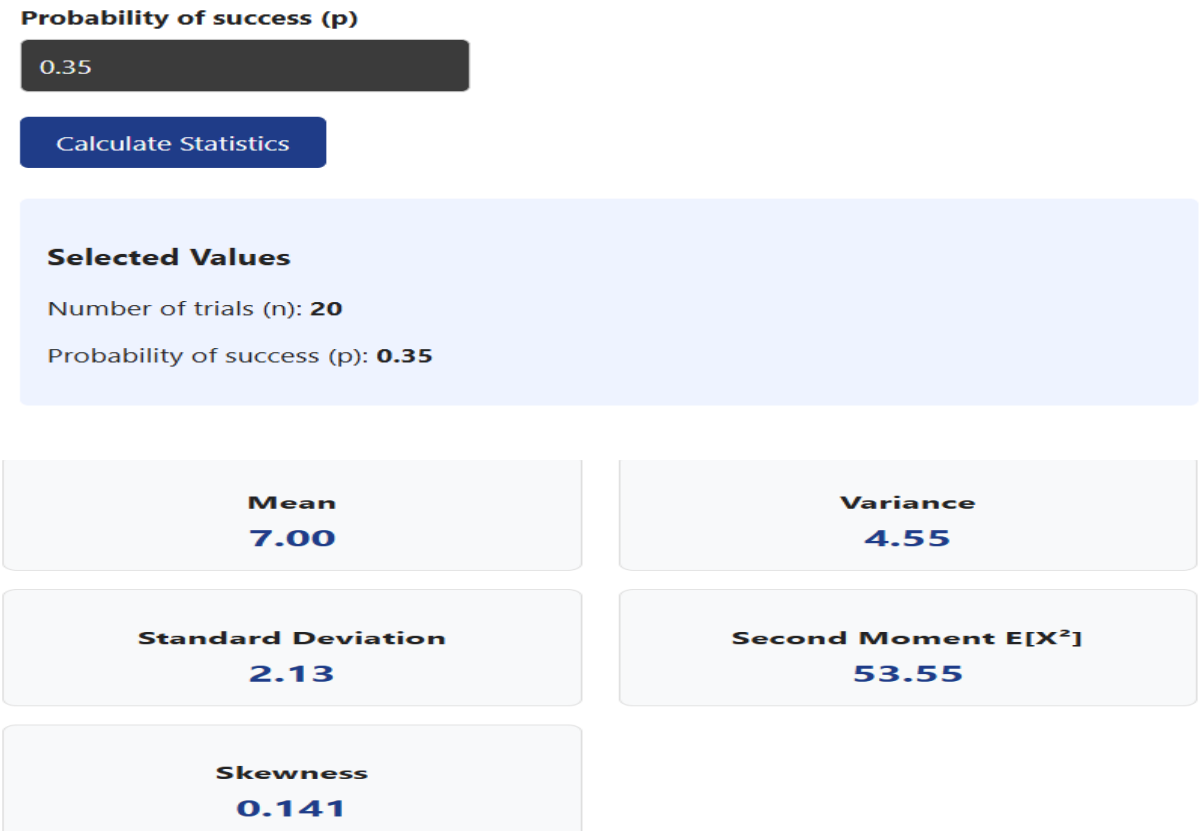
Skewness:

(1-2p) / sqrt(np(1-p))

Display the results in simple statistic cards.

For the default values n=20 and p=0.35, verify that the mean is 7 and variance is 4.55.





Prompt 4 — Add PMF

Add a Binomial PMF calculator.

For:

X ~ Binomial(n,p)

calculate:

P(X = k)

Allow the student to enter k.

Validate that:

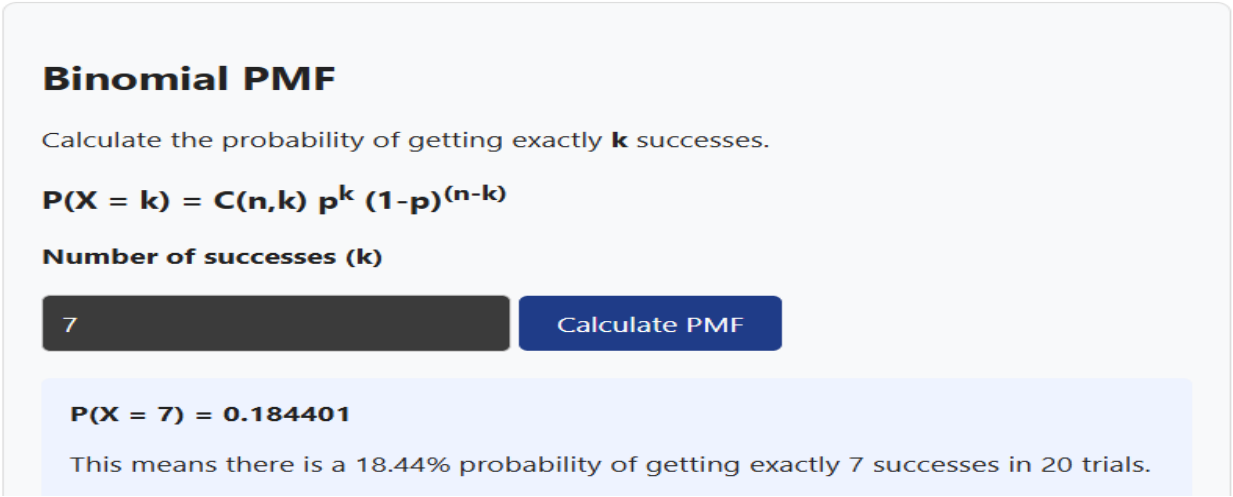
0 <= k <= n

Display the probability clearly.

Also display the mathematical expression:

P(X=k) = C(n,k) p^k (1-p)^(n-k)

Explain in one simple sentence what the result means.



Prompt 5 — Add CDF

Add a Binomial CDF calculator.

Allow the student to enter k.

Calculate:

P(X <= k)

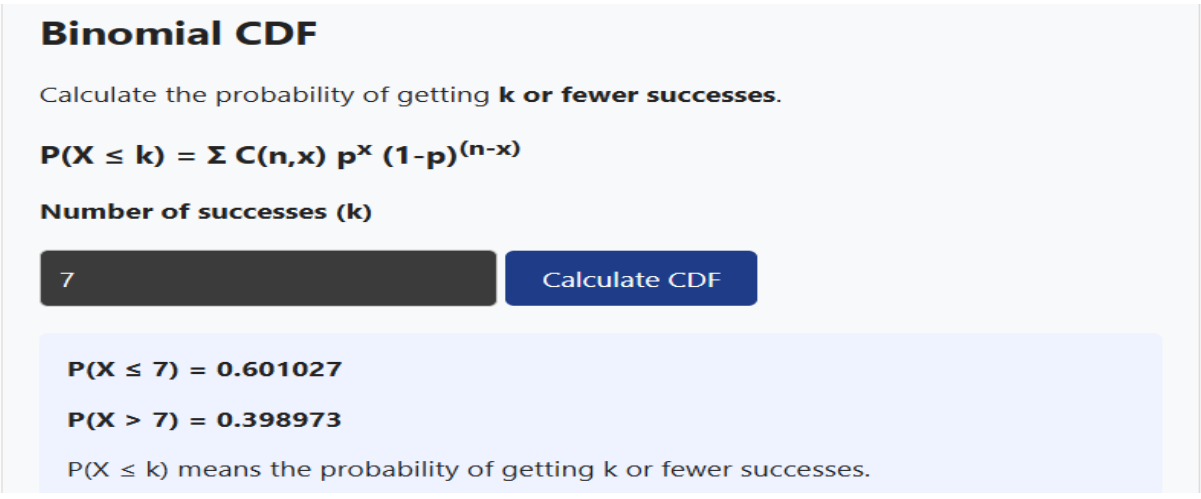
Also calculate:

P(X > k) = 1 - P(X <= k)

Display both results.

Explain:

“P(X <= k) means the probability of getting k or fewer successes.”



Prompt 6 — Add Binomial PMF graph

Add a bar chart showing the Binomial PMF.

X-axis:

Number of successes X

Y-axis:

Probability P(X)

Plot values for:

X = 0, 1, 2, ..., n

Use the current values of n and p.

Add a vertical dashed reference line at the mean.

Add a tooltip showing:

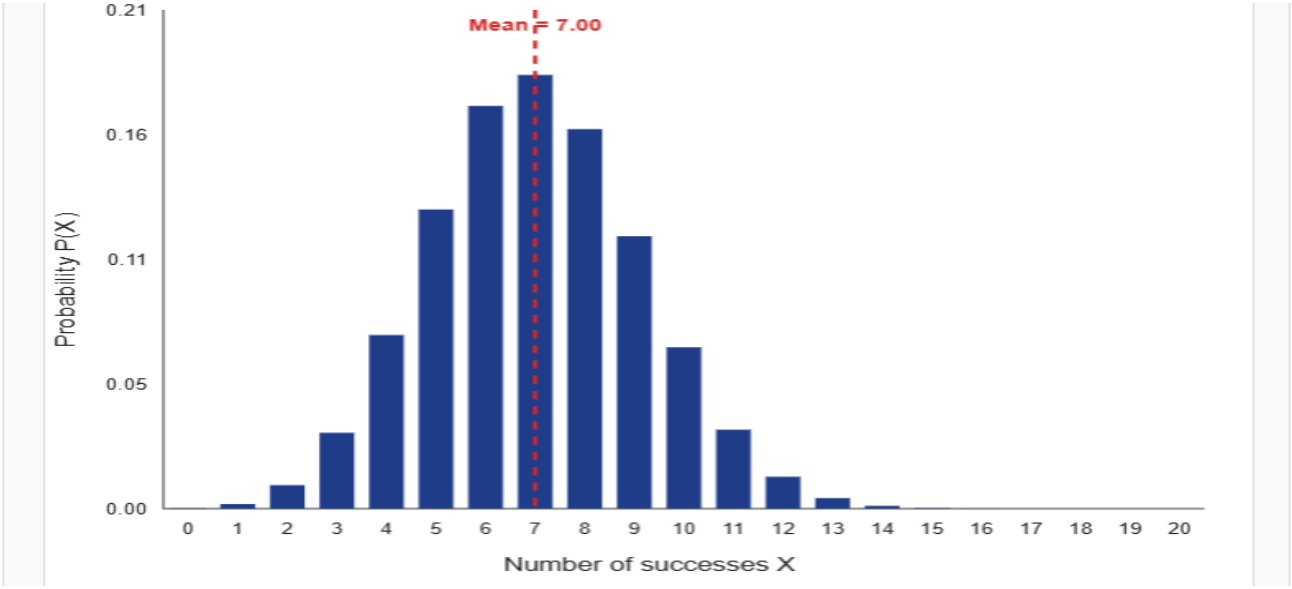
X = value

P(X=value) = probability

Update the graph automatically whenever n or p changes.

Binomial PMF Graph

The graph shows the probability for every possible number of successes from X = 0 to X = n.



Prompt 7:

Add a Poisson Distribution module to my existing “Discrete Probability Distributions” HTML application.

Keep the existing Binomial module fully functional.

Keep the same navigation, design, cards, buttons, and responsive layout.

Add Poisson to the navigation bar.

Set the default rate parameter to $\lambda = 4$.

λ must be greater than 0. Display an error message for invalid inputs.

Add Input Parameters, Statistics, PMF Calculation, CDF Calculation, and a PMF bar chart.

Calculate Mean = λ, Variance = λ, and Standard Deviation = $\sqrt{\lambda}$.

Use PMF: $P(X=k) = (e^{-\lambda} \times \lambda^k) / k!$, where k is a non-negative integer.

Use CDF: $P(X \leq k) = \Sigma\, P(X=i)$, for i = 0 to k.

Also calculate $P(X>k) = 1 - P(X \leq k)$.

Set the default value of k to 4.

Display the results in cards matching the existing Binomial module.

Use purple bars and a vertical dashed red line at the mean in the graph.

When λ changes and the Update button is clicked, automatically update the statistics and graph.

Validate inputs and display appropriate error messages.

Write the actual HTML, CSS, and JavaScript in my existing file. Do not create a separate application or change the existing Binomial module.

Return the complete updated HTML file.

Discrete Probability Distributions

Home | Binomial | Poisson | Geometric

Poisson Distribution

For X ~ Poisson(λ)

Rate parameter λ (lambda)

4

Calculate Statistics

Selected Value:

λ = 4

Mean	Variance	Standard Deviation
4.00	4.00	2.00

Second Moment $E[X^2]$	Skewness
20.00	0.500

Poisson PMF Calculator

For X ~ Poisson(λ)

Enter k

4

Calculate PMF

Mathematical Expression:

$$P(X = k) = (e^{-\lambda} \lambda^k) / k!$$

P(X = k) = **0.195367**

Meaning: The result is the probability of observing exactly **k events** when the average rate of occurrence is **λ**.

Poisson CDF Calculator

For X ~ Poisson(λ)

Enter k

4

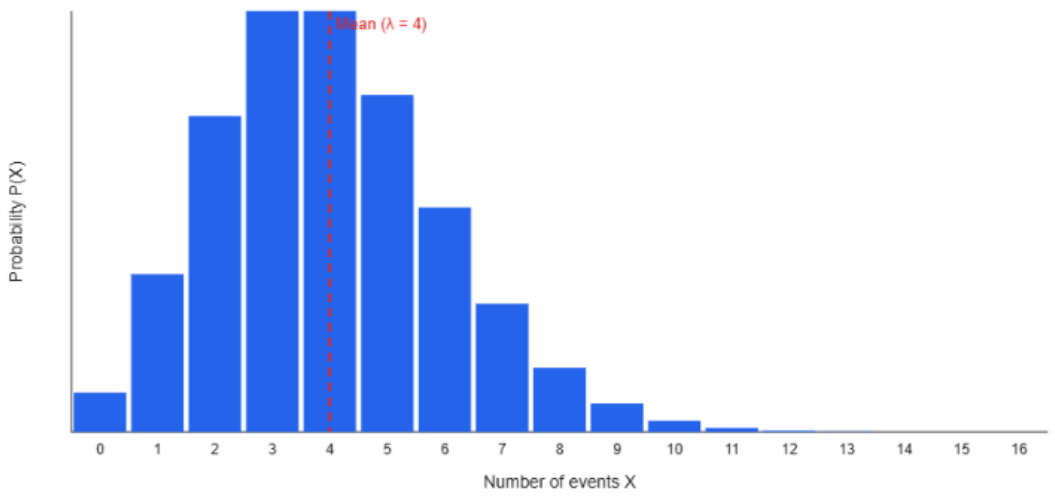
Calculate CDF

P(X ≤ k) = **0.628837**

P(X > k) = **0.371163**

P(X ≤ k) means the probability of getting **k or fewer events.**

Poisson PMF Graph



Prompt 8:

Add a Geometric Distribution module to my existing “Discrete Probability Distributions” HTML application.

Keep the existing Binomial and Poisson modules fully functional.

Keep the same navigation, design, cards, buttons, and responsive layout.

Add Geometric to the navigation bar.

Set the default probability of success to p = 0.35.

Use the convention X = trial number of the first success, starting from 1.

Add Input Parameters, Statistics, PMF Calculation, CDF Calculation, and a PMF bar chart.

Calculate Mean = 1/p, Variance = $(1-p)/p^2$, and Standard Deviation = $\sqrt{(1-p)/p}$.

Use PMF: $P(X=k) = (1-p)^{(k-1)} \times p$.

Use CDF: $P(X \leq k) = 1-(1-p)^k$.

Validate inputs and display error messages.

Use orange bars and a vertical dashed red line at the mean in the graph.

Write the actual HTML, CSS, and JavaScript in my existing file. Do not create a separate application or change the existing modules.

Return the complete updated HTML file.



Geometric Distribution

The Geometric distribution calculates the probability that the first success occurs on a particular trial.

$$X \sim \text{Geometric}(p)$$

p is the probability of success on each independent trial.

Every trial has the same probability of success, and trials are independent.

Convention: X = 1, 2, 3, ...

X represents the trial number on which the first success occurs.

Input Parameter

Probability of success (p)

0.35

Valid probability.

Selected Value

Probability of success (p) = 0.35

Geometric Distribution Statistics

Mean

2.8571

$E[X] = 1/p$

Variance

5.3061

$(1-p)/p^2$

Standard Deviation

2.3035

$\sqrt{(1-p)/p}$

Second Moment $E[X^2]$

13.4694

$Var(X) + (E[X])^2$

Skewness

2.0466

$(2-p)/\sqrt{(1-p)}$

Geometric PMF Calculator

Calculate the probability that the first success occurs on exactly trial x.

$$P(X=x) = (1-p)^{x-1}p$$

Trial number (x)

1

Valid inputs.

PMF Result

$$P(X=x) = (1-p)^{x-1}p$$

Substitution: $P(X=1) = (1-0.35)^{1-1} \times 0.35$

P(X=1) = 0.350000

P(X = x) means the probability that the first success occurs on exactly trial x.

Geometric CDF Calculator

$$P(X \leq x) = 1-(1-p)^{x}$$

$$P(X > x) = (1-p)^{x}$$

CDF Results

Cumulative Probability

$$P(X \leq x) = 1-(1-p)^{x}$$

Substitution: $P(X \leq 1) = 1-(1-0.35)^{1}$

P(X≤1) = 0.350000

P(X ≤ x): The probability that the first success occurs on or before trial x.

Complementary Probability

$$P(X>x) = (1-p)^x$$

Substitution: $P(X>1) = (1-0.35)^1$

$P(X>1) = 0.650000$

P(X > x): The probability that the first success occurs after trial x.

Geometric PMF Bar Chart

The graph begins at X = 1. Bar labels show probability values when space permits.

$$P(X=x) = (1-p)^{x-1}p \quad | \quad E[X] = 1/p$$

Selected p = 0.35 | Mean E[X] = 2.8571 | Trials shown: 1 to 11

